

Smart Home Energy Advisor Agent

Electrical Engineering — Problem Statement No. 16

The Challenge

Rising electricity demand and energy costs make it difficult for households to understand how their daily habits impact power consumption. Most users only receive monthly bills, which provide limited insight into appliance-level energy usage. As a result, people may waste electricity unknowingly. An AI-powered system can analyze usage patterns and provide clear, actionable recommendations for reducing energy consumption and cost.

The Objective

Build an AI-powered Smart Home Energy Advisor Agent that analyzes household electricity consumption data and provides intelligent, personalized recommendations to optimize energy usage.

Core Functionalities

Feature	Description
Energy Consumption Analysis	Analyze household electricity usage patterns from smart meters or appliance usage logs.
Personalized Energy Recommendation	Suggest energy-saving actions based on the user's consumption behavior and daily routines.
Appliance Usage Insights	Identify appliances that consume the most electricity and recommend optimal usage strategies.
Electricity Bill Estimation	Predict monthly electricity bills based on current usage patterns and applicable tariff structures.
Energy Efficiency Suggestions	Recommend practical actions such as reducing peak-hour usage, scheduling heavy appliances.

Expected Agent Workflow

- Collect electricity consumption data from smart meters, appliance logs, or user-provided records.
- Process and analyze consumption by appliance, time period, and usage pattern.
- Use IBM models to interpret patterns and generate personalized recommendations.
- Estimate the user's monthly electricity bill using current consumption and tariff information.
- Present prioritized, easy-to-follow actions that can reduce electricity usage and cost.

Tech Stack (Mandatory)

IBM Langflow / IBM Orchestrate + IBM Models

The solution must use IBM Langflow or IBM Orchestrate for agent/workflow development together with IBM Models for the AI-powered analysis and recommendation layer.

Potential Output to the User

- Daily and monthly energy-consumption summaries.
- Top energy-consuming appliances.
- Estimated monthly electricity bill.

- Personalized energy-saving recommendations.
- Suggestions for shifting heavy appliance usage away from peak hours.
- Long-term energy-efficiency recommendations.

Goal: Help households understand where their electricity is being consumed, predict likely costs, and take simple actions to reduce unnecessary energy usage.